

Multi-Server Java Application Build, Store, and Deployment System

Objective

The objective of this task is to set up a multi-server environment for building, storing, and deploying a Java web application using a Build Server, Nexus Repository Server, and Deploy Server.

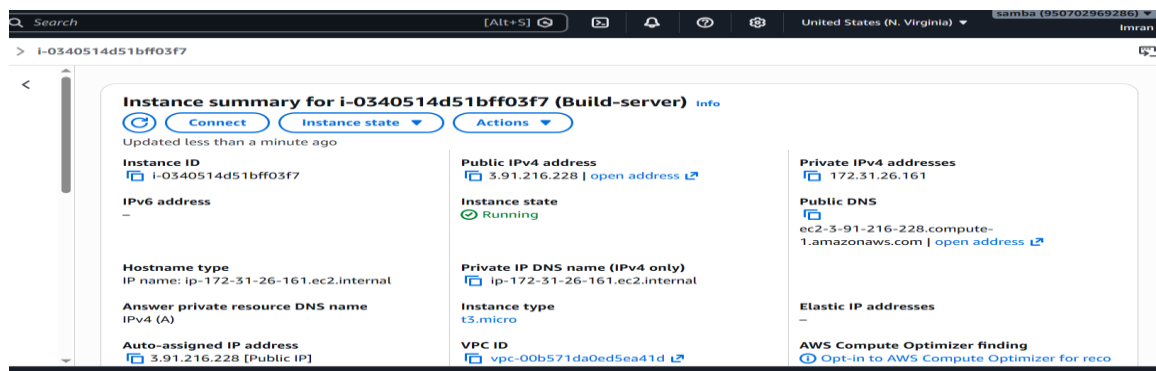
Servers Used

1. Build Server – Build Java application using Maven
2. Nexus Repository Server – Store WAR artifacts
3. Deploy Server – Deploy application on Apache Tomcat

1. Java Build Server Setup

Step 1: Connect to Build Server

```
ssh -i "devops.pem" ubuntu@ec2-3-91-216-228.compute-1.amazonaws.com
```



Step 2: Install Java and Maven

```
sudo apt update
```

```
sudo apt install openjdk-17-jdk -y
```

```
sudo apt install maven -y
```

```
java -version
```

```
mvn -version
```

```
ubuntu@Build-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Build-server:~$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 17.0.18, vendor: Ubuntu, runtime: /usr/lib/jvm/java-17-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-1004-aws", arch: "amd64", family: "unix"
ubuntu@Build-server:~$
```

Step 3: Clone GitHub Repository

```
git clone https://github.com/mrtechreddy/Java-Web-Calculator-App.git
cd Java-Web-Calculator-App
```

```
ubuntu@Build-server:~$ ls
Java-Web-Calculator-App  devops.pem
ubuntu@Build-server:~$ |
```

Step 4: Build Application

mvn clean package

```
[INFO]
[INFO] --- war:3.3.2:war (default-war) @ webapp-add ---
[INFO] Packaging webapp
[INFO] Assembling webapp [webapp-add] in [/home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/ubuntu/Java-Web-Calculator-App/src/main/webapp]
[INFO] Building war: /home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3.war
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.614 s
[INFO] Finished at: 2026-05-17T10:00:48Z
[INFO]
```

WAR file generated inside target/ directory.

```
ubuntu@Build-server:~$ cd Java-Web-Calculator-App/
ubuntu@Build-server:~/Java-Web-Calculator-App$ ls
Jenkinsfile README.md pom.xml src target
ubuntu@Build-server:~/Java-Web-Calculator-App$ cd target/
ubuntu@Build-server:~/Java-Web-Calculator-App/target$ ls
classes          generated-test-sources  maven-status  test-classes  webapp-add-0.0.3.war
generated-sources maven-archiver         surefire-reports webapp-add-0.0.3
ubuntu@Build-server:~/Java-Web-Calculator-App/target$ |
```

Step 5: Configure pom.xml

Add Nexus repository configuration under distributionManagement.

```
<distributionManagement>
  <repository>
    <id>maven-releases</id>
    <url>http://3.80.155.124:8081/repository/maven-releases/</url>
  </repository>
</distributionManagement>
</project>
```

Step 6: Configure Maven settings.xml

Add Nexus credentials in /etc/maven/settings.xml.

```
<!-- Nexus Authentication -->
<servers>

    <server>
<id>maven-releases</id>
<username>admin</username>
<password>admin123</password>
</server>

    <server>
<id>maven-snapshots</id>
<username>admin</username>
<password>admin</password>
</server>

</servers>

<!-- Mirrors -->
<mirrors>
```

Step 7: Deploy Artifact to Nexus

mvn deploy

```
[INFO]
[INFO] --- war:3.3.2:war (default-war) @ webapp-add ---
[INFO] Packaging webapp
[INFO] Assembling webapp [webapp-add] in [/home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/ubuntu/Java-Web-Calculator-App/src/main/webapp]
[INFO] Building war: /home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3.war
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.614 s
[INFO] Finished at: 2026-05-17T10:00:48Z
[INFO]
```

2. Nexus Repository Server Setup

Step 1: Install Java

sudo apt update

sudo apt install openjdk-17-jdk -y

The screenshot shows the AWS Management Console interface. At the top, there's a search bar and navigation icons. Below that, the breadcrumb navigation shows 'ncs > i-Oe83496df2823d81d'. The main content area displays the 'Instance summary for i-Oe83496df2823d81d (Nexus-server)'. There are three buttons: 'Connect', 'Instance state', and 'Actions'. Below these, it says 'Updated less than a minute ago'. The instance details are organized into three columns: 'Instance ID' (i-Oe83496df2823d81d), 'Public IPv4 address' (3.80.59.36), and 'Private IPv4 addresses' (172.31.20.230). The 'Instance state' is 'Running'. The 'Public DNS' is 'ec2-3-80-59-36.compute-1.amazonaws.com'.

```
ubuntu@Nexus-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Nexus-server:~$ |
```

Step 2: Download Nexus

```
14 wget https://download.sonatype.com/nexus/3/nexus-3.85.0-03-linux-x86_64.tar.gz
```

```
ubuntu@Nexus-server:~$ ls
nexus-3.85.0-03  nexus-3.85.0-03-linux-x86_64.tar.gz  sonatype-work
ubuntu@Nexus-server:~$ |
```

Step 3: Extract Nexus

tar -xvf latest-unix.tar.gz

Start Nexus

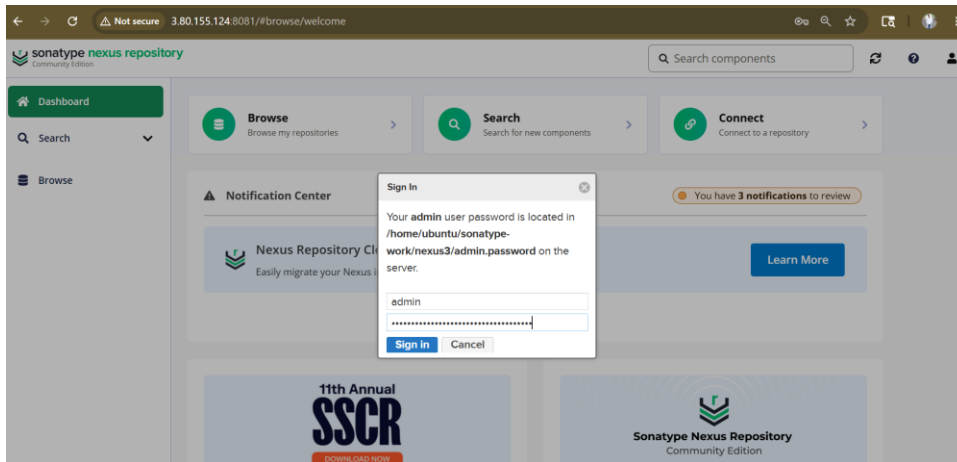
cd /opt/nexus/bin

./nexus start

```
16 tar -xvf nexus-3.85.0-03-linux-x86_64.tar.gz
17 ls
18 cd nexus-3.85.0-03/
19 ls
20 cd bin
21 ls
22 ./nexus
23 ./nexus start
24 ./nexus status
25 ./nexus status
```

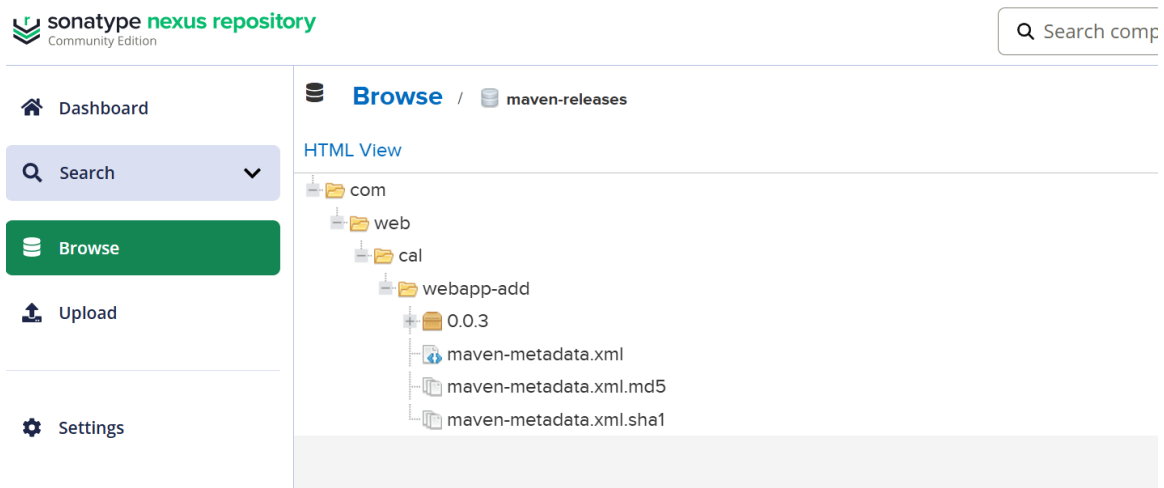
Step 5: Access Nexus Web Interface

http://3.80.59.36:8081



Step 6: Create Maven Hosted Repository

Create repository named maven-releases.

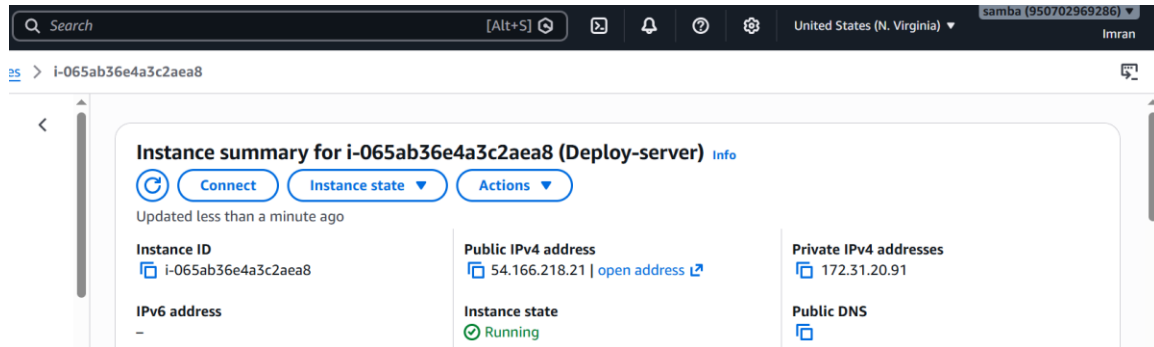


3. Deploy Server Setup

Step 1: Install Java and Apache Tomcat

```
sudo apt update
```

```
sudo apt install openjdk-17-jdk -y
```



```
ubuntu@Deploy-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Deploy-server:~$
```

Download Tomcat:

```
wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz
```

```
8 wget https://dldn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz
9 ls
10 tar -xvf apache-tomcat-9.0.118.tar.gz
```

Extract Tomcat:

```
tar -xvf apache-tomcat-9.0.118.tar.gz
```

```
apache-tomcat-9.0.118  apache-tomcat-9.0.118.tar.gz
ubuntu@Deploy-server:~$
```

Step 2: Start Tomcat

cd /opt/tomcat/bin

./startup.sh

```
ubuntu@Deploy-server:~$ cd apache-tomcat-9.0.118/
ubuntu@Deploy-server:~/apache-tomcat-9.0.118$ ls
BUILDING.txt  LICENSE  README.md  RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE  RELEASE-NOTES  bin  lib  temp  work
ubuntu@Deploy-server:~/apache-tomcat-9.0.118$ cd bin
ubuntu@Deploy-server:~/apache-tomcat-9.0.118/bin$ ls
bootstrap.jar  ciphers.sh  commons-daemon-native.tar.gz  daemon.sh  setclasspath.bat  startup.sh  version
catalina-tasks.xml  commons-daemon.jar  digest.bat  setclasspath.sh  tomcat-juli.jar  version
catalina.bat  configtest.bat  makebase.bat  shutdown.bat  tomcat-native.tar.gz
ciphers.bat  configtest.sh  makebase.sh  startup.bat  tool-wrapper.bat
tool-wrapper.sh
ubuntu@Deploy-server:~/apache-tomcat-9.0.118/bin$ ./startup.sh
Using CATALINA_BASE:   /home/ubuntu/apache-tomcat-9.0.118
Using CATALINA_HOME:   /home/ubuntu/apache-tomcat-9.0.118
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.118/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ubuntu/apache-tomcat-9.0.118/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.118/b
t-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

⚠ Not secure 54.166.218.21:8080

Home Documentation Configuration Examples Wiki Mailing Lists

Apache Tomcat/9.0.118

If you're seeing this, you've successfully installed Tomcat. Co



Recommended Reading:
[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

Developer Quick Start
[Tomcat Setup](#) [Realms & AAA](#) [Examples](#)
[First Web Application](#) [JDBC DataSources](#)

Step 3: Copy WAR File to Deploy Server

```
126 scp -i /home/ubuntu/devops.pem Java-Web-Calculator-App/target/webapp-add-0.0.3.war ubuntu@54.160.139.225:/home/ub
rtu/apache-tomcat-9.0.118/webapps/
```

Step 4: Access Application
http://54.166.218.21:8080

Addition

5

Calculator

first number:

Second number :

☒ addition
☐ subtraction
☐ production

Issues Encountered and Fixes

1. Permission denied for PEM file → Fixed using `chmod 400 devops.pem`
2. Maven deployment failed → Corrected Nexus repository ID and credentials
3. Tomcat startup issue → Verified Java installation and Tomcat logs
4. SCP connection issue → Updated security groups and PEM permissions

Conclusion

Successfully implemented a complete multi-server Java application deployment pipeline using Apache Maven, Nexus Repository Manager, and Apache Tomcat.